# Bayesian Time-Varying Multi-Seasonal ARMA Models

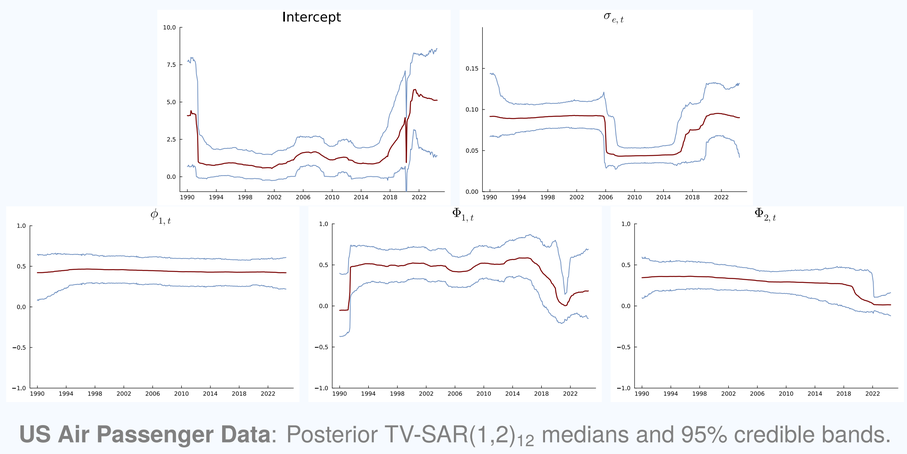
## Robust, Stable and Scalable Inference for Nonstationary Time Series

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## 1. Why TV-SARMA Models?

**Static Model (Box & Jenkins)** $\Longrightarrow$ **TV Model (Box & Jenkins 2.0)**

$$\phi(L)\Phi(L^s)y_t = \beta_0 + \psi(L)\Psi(L^s)\varepsilon_t \qquad \phi_t(L)\Phi_t(L^s)y_t = \beta_{0,t} + \psi_t(L)\Psi_t(L^s)\varepsilon_t$$




**US Air Passenger Data**: Posterior TV-SAR(1,2)$_{12}$ medians and 95% credible bands.

TV-SAR captures how major events, such as the COVID-19 pandemic, alter trend, seasonality and volatility over time. In contrast, static models assume fixed dynamics.

## 2. Modeling Framework and Contributions

**Bayesian** | **TV-SARMA** | **Multiple Seasons** | **Stable & Invertibile**
**DSP Priors** | **Nonlinear Inference** | **Adaptive Filtering** | **Scalable**

## 3. State-Space Representation

$$\textbf{Observation: } y_t = \boldsymbol{x}_{0t}^\top \tilde{\boldsymbol{\phi}}_t^{(AR)} + \boldsymbol{z}_{0t}^\top \tilde{\boldsymbol{\psi}}_t^{(MA)} + \varepsilon_t, \qquad \varepsilon_t \sim N(0,\sigma^2)$$

$$\textbf{Constraint: } \tilde{\boldsymbol{\phi}}_t, \tilde{\boldsymbol{\psi}}_t = \tilde{\boldsymbol{g}}(\boldsymbol{\theta}_t) \qquad \textbf{(stability \& invertibility)}$$

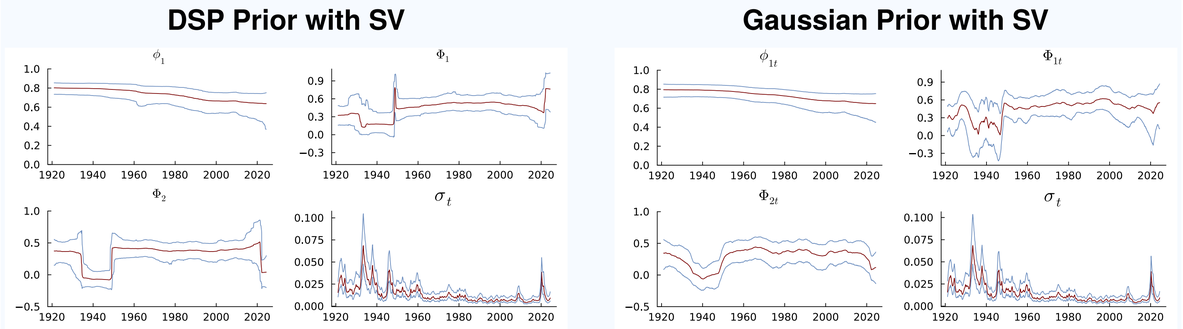
$$\textbf{State: } \boldsymbol{\theta}_t = \boldsymbol{\theta}_{t-1} + \boldsymbol{\nu}_t, \qquad \boldsymbol{\nu}_t \sim N(0, \mathrm{Diag}\{\exp(\boldsymbol{h}_t)\})$$

$$\textbf{DSP: } \boldsymbol{h}_t = \boldsymbol{\mu} + \boldsymbol{\kappa}(\boldsymbol{h}_{t-1} - \boldsymbol{\mu}) + \boldsymbol{\eta}_t, \qquad \boldsymbol{\eta}_t \sim Z(1/2, 1/2, 0, 1).$$

The observation equation is the multiplicative SARMA model in nonlinear regression form. Here, $\boldsymbol{x}_{0t}$ and $\boldsymbol{z}_{0t}$ collect the nonzero lagged observations and innovations (including presample and initial values), while $\tilde{\phi}_t$ and $\tilde{\psi}_t$ contain the corresponding nonzero AR and MA coefficients, including interaction terms.

The formulation naturally accommodates multiple seasonal periods.

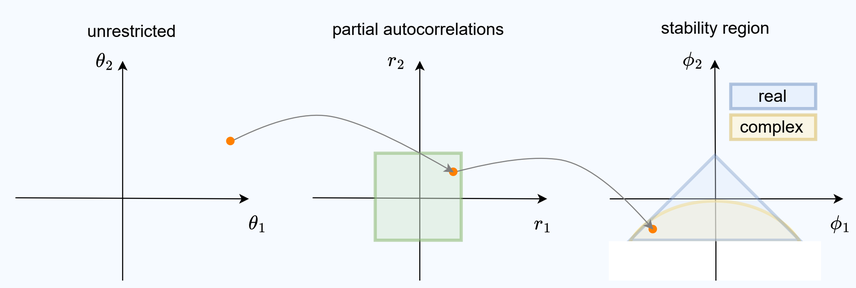
## 4. Dynamic Shrinkage vs. Gaussian Regularization




**US Industry Production Index Data**: Posterior TV-SAR(1,2)$_{12}$ medians and 95% credible bands.

**DSP prior** (left): adaptive shrinkage, smoother trajectories, and abrupt-change detection.
**Gaussian prior** (right): less regularization, more variable trajectories.

## 5. Stability Parametrization and Uniform Prior



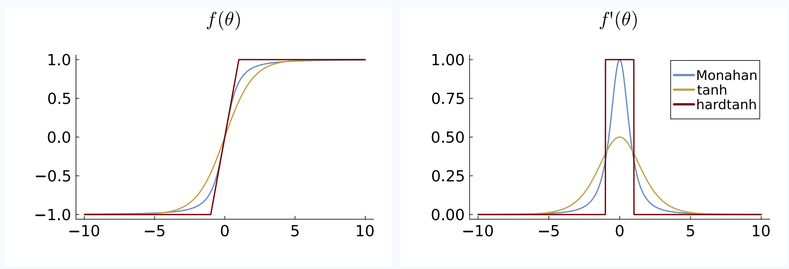

$$\theta \xrightarrow{\text{Monahan}} r \xrightarrow{\text{Levinson–Durbin}} \phi$$

**Uniform prior over the stable AR region with Monahan (1984)**

Same idea applies to ensure invertibility!

## 6. Robust Stability Parametrization

**Monahan (1984)** $\Longrightarrow$ tanh $\Longrightarrow$ **hardtanh**

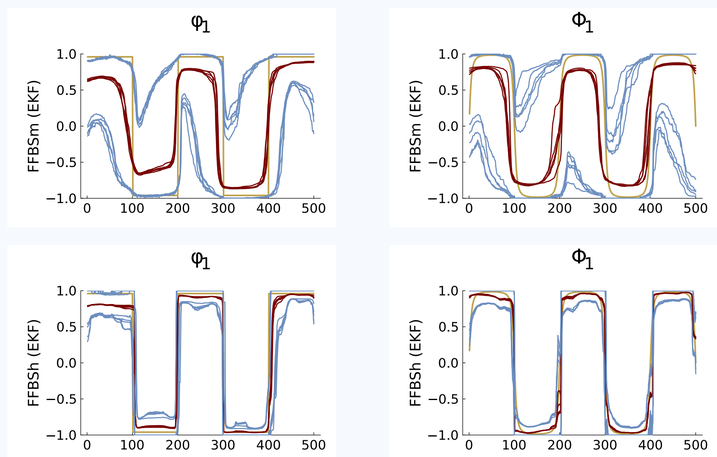



$$\frac{\theta}{\sqrt{1+\theta^2}} \longrightarrow \tanh(\theta/2) \longrightarrow \min(1, \max(-1, \theta))$$

(*vanishing gradients*) (*vanishing gradients*) (***linear interior***)

## 7. Near the Stability Boundary

**TV-SAR(1,1)$_{12}$** | **Extreme Boundary Dynamics** | **Simulated Data**



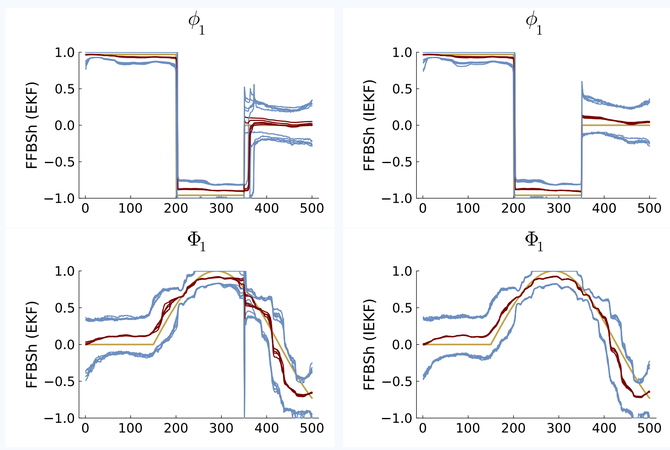

Posterior medians and 95% credible bands from five independent runs under **Monahan** (first row) and **hardtanh** (second row) parametrizations.

**Monahan** (first row): unstable filtering and inflated credible bands.
**Hardtanh + constrained filtering** (second row): robust, numerically stable inference.

## 8. Robust Adaptive Inference

**Hardtanh** | **Constrained Filtering** | **Adaptive Inference**

**FFBS+EKF** (*Fast*) $\xrightarrow{\text{NIS trigger}}$ **FFBS+IEKF** (*Accurate*)




**Simulated TV-SAR**$(1,1)_{12}$: posterior medians and 95% credible bands from five independent runs under **FFBS+EKF** (left column) and **FFBS+IEKF** (right column).

**Adaptive inference:** fast when possible, accurate when necessary!

## 9. Gibbs sampling scheme

**States** $\Longrightarrow$ **DSP** $\Longrightarrow$ **Presample** $\Longrightarrow$ **Errors** $\Longrightarrow$ **Volatility**

(FFBS+(I)EKF) (Kowal, 2017) (Closed-form) (Closed-form) (SV+DSP)

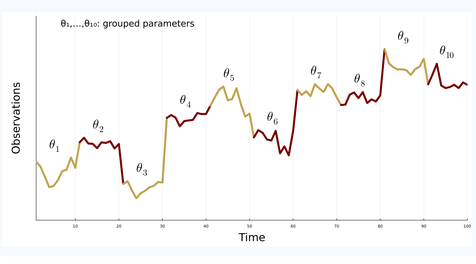
## 10. Scaling to Multiple Seasonalities

**Grouping** $\Longrightarrow$ **Sequential EKF** $\Longrightarrow$ **Scalable Inference**

(*Fewer state updates*) (*Lower filtering cost*) (*Long high-frequency series*)

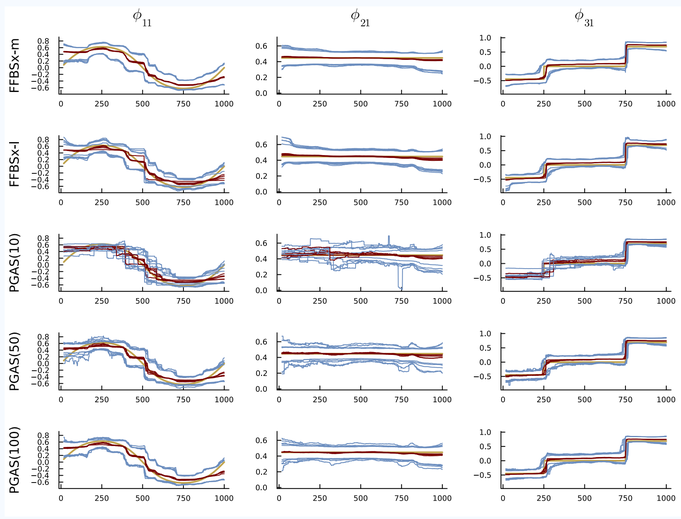
**Grouped Computation** $\Longrightarrow$ **Prior-Preserving**

(Faster inference) (Group-adjusted innovations)




Grouping is a computational device, not a modeling assumption. Observations and latent states evolve on different time scales. Group-adjusted innovations preserve the original state dynamics. Faster inference.

## 11. Simulations




**TV-SAR(1,1,1)$_{(4,12)}$**: Posterior medians (red) and 95% credible bands (blue) from five independent runs.

**Poster** **Paper** **Code**

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